Advanced Square Roots of Non-Perfect Squares – Worksheet Set 2

SQR = square root

1. Simplify completely: √128. 8 sqr2

2. Simplify and combine like radicals: 3√50 – 2√18 + √8. 13 sqr2

3. Rationalize the denominator: 9 / √13. sqr13/117

4. Find two consecutive integers between which √70 lies. 8,9

5. Evaluate: (2√3 + √12)(√6). 8.6 sqr3

6. Simplify: (√15 – √5)². 170

7. Rationalize the denominator: 6 / (√5 – 1). 7.2/3.6

8. A rectangular garden measures √45 m by √80 m. Find its exact area. 6.6 by 8.8

9. Solve for x: √(5x – 4) = 6. x=8

10. Simplify completely: √147 + √75 – √48. 8 sqr3

11. The area of a square is 162 cm². Find the exact side length. 12.6 to 1 dp

12. Evaluate without a calculator: √32 × √18. 9.8 to 1 dp

13. Rationalize and simplify: (3√2) / (√6 + √3). 3 sqr2/ sqr19

14. A ladder 13 m long leans against a wall. If the base is 5 m from the wall, find the exact height reached by the ladder. 8 meters

15. Simplify: (√27 + √12) ÷ √3. 5 sqr3/sqr3 or 5